

MEDICAL IMAGE CLASSIFICATION OF CT SCANS USING U-NET FOR TUMOR DIAGNOSIS

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ABSTRACT

This project presents a deep learning pipeline for brain tumor identification that integrates segmentation and classification using a modified U-Net architecture. Leveraging multimodal imaging data from a Kaggle dataset (1496 images; 26 misclassifications), the model achieved an overall accuracy of 98.26% (Healthy: 98.35%, Tumor: 98.19%).

The approach produces both pixel-wise segmentation maps and tumor probability scores, offering a comprehensive tool for automated diagnosis. The code and full implementation are available at GitHub.

1. INTRODUCTION

Early detection of brain tumors is critical for effective treatment and improved patient outcomes, as timely diagnosis can significantly enhance the chances of successful intervention. Recent advances in deep learning have demonstrated remarkable potential in the field of medical image analysis [5][6], particularly for tasks such as image segmentation and classification [4][8]. These technologies offer a robust alternative to traditional methods, providing faster, more consistent, and highly accurate analyses that are essential in clinical environments where rapid decision-making is paramount.

In this project, we focus on harnessing the power of the U-Net architecture—a deep convolutional neural network that has been widely adopted in the segmentation of biomedical images—to tackle the challenges associated with brain tumor detection. U-Net and its various derivatives have been particularly effective in delineating tumor boundaries in both MRI and CT scans [9], thanks to their unique encoder-decoder structure that captures contextual information and accurately localizes regions of interest.

2. RELATED WORK

Several studies have explored deep learning for brain tumor detection, with U-Net emerging as a powerful tool for medical image segmentation due to its ability to capture fine details and accurately localize tumor regions [3][8]. The encoder-decoder structure of U-Net allows it to excel in biomedical image segmentation, including brain tumor delineation from MRI and CT scans. This architecture has become a standard approach due to its efficiency in learning both global and local features within medical images.

In addition to U-Net, various convolutional neural networks (CNNs) and their variants have been employed for tumor classification. López and Choi [1] introduced DeepSeg, which combines multiple CNN architectures within a modified U-Net to improve segmentation performance in brain tumor detection. Their work demonstrates how the integration of different architectures can enhance tumor localization. Similarly, Abd-Ellah et al. [2] proposed a cascaded deep CNN model that incorporates both symmetric U-Net and asymmetric residual blocks, achieving improved classification and segmentation performance for brain tumors.

Other research has demonstrated the potential of combining U-Net with advanced deep learning models, such as ResNet. Asiri et al. [7] explored the use of fine-tuned CNNs with ResNet50 and U-Net for brain tumor detection and classification, showcasing the benefits of this hybrid model for MRI-based applications. These advancements highlight the growing importance of deep learning models in both segmentation and classification tasks for brain tumor analysis.

Additionally, Sun et al. [10] proposed an improved FPN (Feature Pyramid Network) for brain tumor segmentation, further advancing the field by incorporating multi-scale feature learning. Similarly, Aggarwal et al. [11] focused on early detection and segmentation of brain tumors using deep neural networks, emphasizing the importance of precise tumor localization for early diagnosis.

Inspired by these developments, this project integrates both segmentation and classification into a unified framework using a modified U-Net architecture, with the goal of enhancing tumor detection accuracy. Our approach builds on the strengths of existing research while also addressing the challenges of limited supervision for segmentation in certain datasets.

3. MATERIALS AND METHODS

3.1 Dataset and Preprocessing

The dataset utilized in this project is sourced from Kaggle. It contains 4620 images acquired from multimodal CT and 5002 images acquired from MRI scans. For our purposes, we will be using 2D CT scans. The 2D CT scans dataset contains 2301 healthy-case images and 2319 tumor-bearing images. Notably, the dataset provides only image-level labels (healthy vs. tumor) without corresponding segmentation masks. Therefore, while a segmentation branch was incorporated into the network as a proof-of-concept, it was not trained with reliable supervision. In the training stage, all tumor image-label images are manually assigned with a masking score of 0.5, while healthy-label images are assigned with a score of 0.

The image preprocessing workflow begins by reading each image using OpenCV and resizing them to 256×256 pixels. Along with the resized images, the filename and corresponding label are stored. The dataset is then randomly shuffled and partitioned into three subsets: 75% for training, 12.5% for validation, and 12.5% for testing. Each image is subsequently saved in an appropriate directory structure—organized under three main directories (train, val, and test) with separate subfolders for healthy and tumor images—while retaining its original filename.

3.2 Model Architecture

Our deep learning model, implemented using PyTorch, is structured around an encoder–bridge–decoder architecture augmented with a dedicated classification branch (see Figure 1). The model accepts RGB images (3 channels) as input. The encoder comprises four convolutional blocks, each containing two consecutive 3×3 convolutions accompanied by batch normalization and ReLU activation. With each block, the number of filters is doubled, while MaxPooling layers are applied to reduce the spatial dimensions by half. When data arrives at the bridge, it has size of 16×16 with 1024 channels. The encoder-decoder architecture, particularly in its U-Net form, is widely used for biomedical image segmentation due to its ability to capture both global and local features [12].

After the bridge, the decoder mirrors this design with four up-sampling blocks. Each block utilizes a transposed convolution followed by a convolutional block similar to those in the encoder, ultimately producing a segmentation output through a single channel. Parallel to the decoder, a classification branch is integrated into the model. This branch takes the output directly from the bridge which contains the most compressed but feature-rich representation then uses global average pooling (GAP) followed by 2 fully connected (FC) layers to generate the final classification output.

3.3 Forward Pass

During inference, the input images first traverse the encoder blocks before passing through a central bridge. The subsequent decoder phase, augmented with skip connections from the encoder, refines the segmentation prediction. In parallel, features from the bridge are channeled into the classification branch, which robustly determines whether the image depicts a healthy or tumor-bearing case.

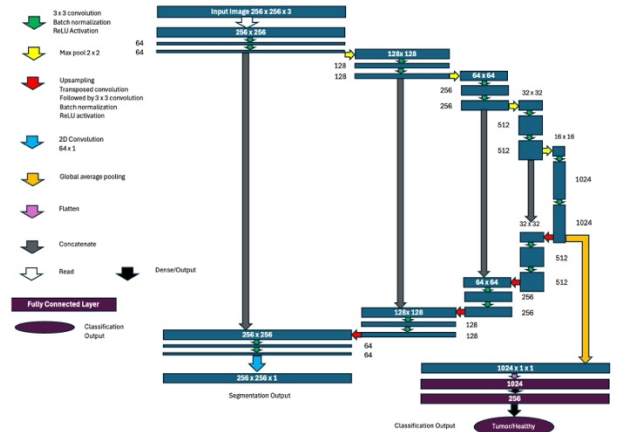


Figure 1. Brain tumor classification using U-Net Model

4. RESULTS AND DISCUSSION

Our integrated U-Net model demonstrates exceptional performance on the Kaggle dataset for automated brain tumor diagnosis. The model achieves an overall accuracy of **98.11%**, with healthy images classified at **96.80%** accuracy and tumor-bearing images at **99.09%**. Out of 581 images (12.5% of CT scans), only 11 instances were misclassified, demonstrating the model's robust capability for distinguishing between healthy and tumor cases (see Figure 2 for the confusion matrix).

Although we incorporated a segmentation branch with the aim of exploring tumor localization, its performance could not be thoroughly assessed due to the lack of annotated segmentation masks in the dataset. As such, our evaluation has focused primarily on classification metrics. This decision is further supported by qualitative analyses, including a detailed examination of misclassified examples (Figure 2) and representative instances of correctly classified images (Figure 3). These supplementary figures not only validate the statistical performance but also provide insight into the model's behavior under various conditions.

While the high accuracies illustrate the potential for real-world application, certain limitations must be acknowledged. The modest size of the dataset may affect the generalizability of our findings. Future work will involve the use of larger, more diverse datasets, the incorporation of additional imaging modalities, and the exploration of more advanced architectures. Additionally, integrating datasets with comprehensive segmentation annotations will allow for a more complete exploitation of the model's dual capabilities in both segmentation and classification.

The source code of this project can be found on: https://github.com/ENEL-645-group6/645_final

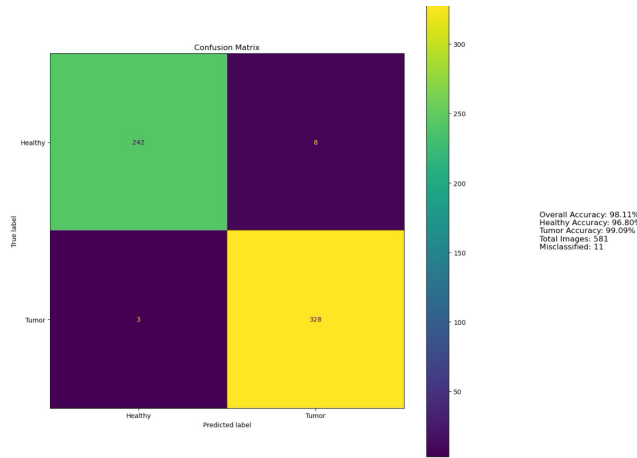


Figure 2. Confusion Matrix

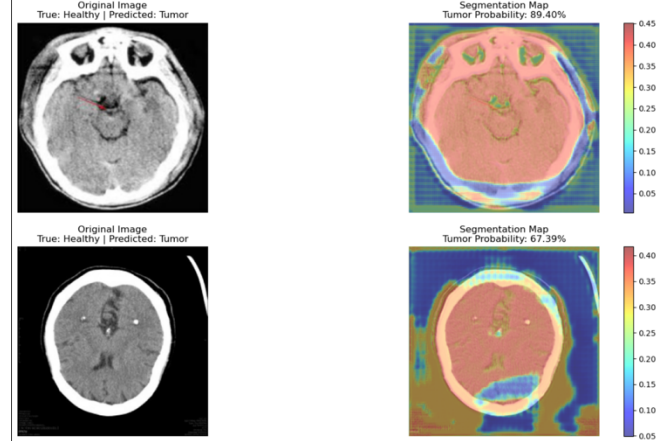


Figure 3. Misclassified examples

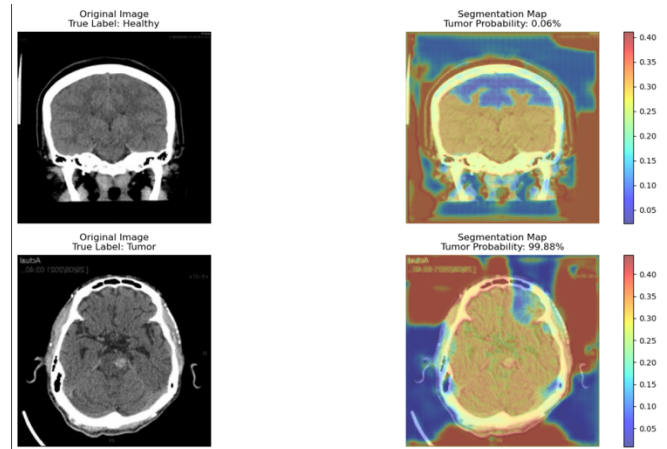


Figure 4. Correctly classified examples

5. CONCLUSION

This project demonstrates the successful application of a U-Net architecture for brain tumor detection, achieving high classification accuracy despite the challenges imposed by limited segmentation supervision. The integrated framework effectively distinguishes between healthy and tumor-bearing images, underscoring the potential of deep learning approaches for automated brain tumor diagnosis. While the segmentation branch was incorporated as a conceptual extension to explore tumor localization, its evaluation was constrained by the absence of ground truth segmentation masks. Nevertheless, the robust performance of the classification branch indicates that the model can serve as a reliable diagnostic tool.

Our work highlights several key contributions. First, it validates the feasibility of using U-Net-based architectures for medical image analysis in the context of brain tumor detection. Second, it presents a unified approach that simultaneously addresses both segmentation and classification, paving the way for more comprehensive

diagnostic systems. Third, the project underscores the importance of data quality and diversity, as the current dataset's limitations point to the need for larger, more diverse collections that include detailed segmentation annotations.

Looking ahead, future research will focus on expanding the dataset and incorporating annotated segmentation masks to fully leverage the dual capabilities of the model. Refinements to the segmentation branch will aim to enhance tumor localization, thereby providing more precise diagnostic insights. Additionally, exploring more sophisticated architectures and integrating multimodal imaging data could further improve diagnostic accuracy and generalizability.

14. REFERENCES

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COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by (<https://www.kaggle.com/datasets/murtozalikhon/brain-tumor-multimodal-image-ct-and-mri?select=Dataset>).Ethical approval was not required as confirmed by the license attached with the open access data

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